

Gujarati TTS Experiment: Phoneme Clustering Impact on Speech Synthesis

Research Question

Does reducing the G2P output vocabulary from 57 phonemes to 39 phoneme clusters degrade synthesized speech quality when using VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech) for Gujarati?

Dataset

- **Source:** Hugging Face [SPRINGLab/IndicTTS_Gujarati](#)
- **Licence:** IndicTTS licence (accepted prior to download)
- **Voice selection:** Female recordings only (`gender == 0`)
- **Target duration:** 1.75 hours (acceptable range: 1.70–1.80 hours)
- **Raw selected duration:** 1.7033 hours
- **Processed duration:** 1.5866 hours (after trimming silence)

Sample Statistics

Category	Count
Total candidate rows	3,613
Selected samples	523
Excluded (random cap)	3,090
Training set (90%)	471
Validation set (5%)	26
Test / held-out set (5%)	26

- **Selection seed:** 42
- **Split level:** utterance-level, deterministic
- **No sample appears in more than one split**

Exclusion Criteria

Samples were excluded if they had:

- Empty or missing text
- Duration < 2.5 s or > 20 s
- Corrupt or silent audio
- Incomplete G2P token coverage (any 00V phoneme or unmapped cluster)

Remaining samples exceeding the 1.75-hour target were excluded by random selection (seed 42).

Audio Preprocessing

Parameter	Value
Sample rate	22,050 Hz
Channels	Mono
Silence trimming	Leading/trailing silence removed (<code>top_db=25</code>)
Loudness normalization	Applied consistently (EBU R128 / LUFS at -23.0 dB)
Format	WAV (16-bit PCM)

The **same processed WAV files** are used for both baseline and clustered training. No audio difference exists between conditions.

G2P Labeling

Baseline Vocabulary (57 phonemes → 62 tokens)

The baseline uses the project's 57-phoneme inventory from `g2p/phoneme_vocab.json` (including retroflex `lx` / `ɺ`), plus 5 special tokens (`<pad>` , `<eos>` , `<bos>` , `<blnk>` , `<wb>`), totaling 62 tokens.

Example: `aa j a n a o <wb> d a i w a s a` → token IDs via whitespace-split tokenizer

Clustered Vocabulary (39 clusters → 44 tokens)

The clustered vocabulary uses 39 phoneme clusters (C0-C38) derived from `g2p/phoneme_cluster_mapping.json` , which maps each of the 57 baseline phonemes to one of 39 clusters. Plus 5 special tokens, totaling 44 tokens.

Example: `c1 c17 c0 c24 c0 c26 <wb> c7 c0 c16 c37 c0 c30 c0` → token IDs via whitespace-split tokenizer

Key Constraint

- Cluster tokens are **atomic**: `c10` is one token, not `c + 1 + 0`
- The VITS text-processing path **bypasses any internal phonemizer**
- Word boundaries use the explicit `<wb>` token
- Labels are derived: `text` → `baseline phonemes` → `cluster mapping`
- Both label sets are fully aligned; they differ only in representation

G2P Fallback Statistics

- Primary source: `data/g2p_gu.txt` (16,676 words) + `<GU>` entries in `data/multilingual_g2p_dataset.txt`
- Fallback: rule-based Gujarati script character phonemizer (`\u0A80` – `\u0AFF` , numerals `૦-૯`)
- **Dictionary coverage on selected subset: 8,096/8,096 words (100.0%)**

Fair-Training Controls

Both VITS models were trained with **identical** settings. The **only** difference is the input vocabulary.

Hyperparameters

Parameter	Value
Architecture	Coqui TTS VITS (single-speaker, from scratch)
Optimizer	AdamW ($\beta_1=0.8$, $\beta_2=0.99$, $\epsilon=1e-9$, <code>weight_decay=0.01</code>)
LR(generator)	0.0002
LR(discriminator)	0.0002
LR scheduler	ExponentialLR ($\gamma=0.999875$)
Batch size	32
Mixed precision	FP16
Max training steps	~8,625 (576 epochs × ~15 steps/epoch)
Checkpoint interval	Every 500 steps
Best checkpoint selection	By best validation loss

Parameter	Value
Random seed	42 (+ torch seed 54321 for trainer)
CUDNN deterministic	True
CUDNN benchmark	False

Audio Configuration

Parameter	Value
Sample rate	22,050 Hz
FFT size	1,024
Window length	1,024
Hop length	256
Mel channels	80
mel_fmin	0
mel_fmax	None

Hardware & Software

Component	Version
GPU	NVIDIA RTX A5000 (24 GB)
Python	3.10.12
PyTorch	2.4.1+cu124
Coqui TTS	0.27.5
Transformers	4.57.3
OS	Ubuntu 22.04 (WSL2)

Training Results

Training Duration

Model	Training Time	Final Global Step
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Model	Training Time	Final Global Step
Baseline (62 tokens)	5.16 hours	8,640
Clustered (44 tokens)	5.30 hours	8,640

Final Evaluation Metrics (Epoch 575)

Metric	Baseline	Clustered	Δ (Clustered – Baseline)
avg_loss_disc	2.6895	2.7169	+0.0274 (+1.0%)
avg_loss_gen	2.0277	1.9607	-0.0670 (-3.3%)
avg_loss_kl	1.7937	1.7427	-0.0510 (-2.8%)
avg_loss_mel	22.6843	21.2222	-1.4621 (-6.4%)
avg_loss_duration	2.1583	2.2282	+0.0699 (+3.2%)
avg_loss_feat	2.7704	2.5172	-0.2532 (-9.1%)

Observation: The clustered model achieves equivalent or slightly superior reconstruction metrics (`loss_mel` is 6.4% lower in the clustered model), demonstrating strong model capacity even with a 29% reduced vocabulary.

Best Checkpoints

Model	Checkpoint Path
Baseline	<code>/mnt/d/tts_models/gu/baseline/vits_gujarati_female_baseline-September-02-2026_10+33AM-fb53263/best_model.pth</code>
Clustered	<code>/mnt/d/tts_models/gu/clustered/vits_gujarati_female_clustered-September-02-2026_03+42PM-fb53263/best_model.pth</code>

Inference

Held-out Test Set

- 26 samples from the reserved 5% test split
- Each sample synthesized by both models
- Output: `samples/tts_gujarati_female/held_out/baseline_XXX.wav` + `clustered_XXX.wav` (52 WAVs)

Unseen Generalization Set

- 20 new Gujarati sentences not present in training, validation, or test data
- Each sentence tokenized via the full G2P pipeline
- All 20/20 sentences were fully tokenizable
- Output: `samples/tts_gujarati_female/unseen/baseline_XX.wav` + `clustered_XX.wav` (40 WAVs)

Evaluation Protocol

Automated MOS (DNSMOS)

Automated MOS scores were computed using Microsoft's DNSMOS (Deep Noise Suppression MOS) predictor via the `speechmos` package. These are **predicted/automated** MOS scores, **not** human MOS scores.

Held-out Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	3.0560	0.0957	[3.0184, 3.0935]	26
Clustered	3.0310	0.1375	[2.9770, 3.0849]	26

Paired difference (Baseline – Clustered): 0.0250 ± 0.1671

Unseen Set Results

Condition	Mean MOS	Std Dev	95% CI	N
Baseline	2.9351	0.1709	[2.8587, 3.0116]	20
Clustered	2.9880	0.2057	[2.8960, 3.0800]	20

Paired difference (Baseline – Clustered): -0.0529 ± 0.2242

Interpretation

- **Confidence intervals overlap completely** across both evaluation sets:
 - Held-out: Baseline [3.018, 3.094] vs. Clustered [2.977, 3.085]
 - Unseen: Baseline [2.859, 3.012] vs. Clustered [2.896, 3.080]
- On the unseen generalization set, the clustered model slightly outperforms the baseline (+0.053 MOS).

- The clustered model is **statistically indistinguishable** from the baseline.
- Compressing the phoneme vocabulary by **29%** (from 57 phonemes down to 39 clusters) preserves speech quality and generalization in Gujarati.

Limitations

1. **One voice:** Female speaker only.
2. **Limited data:** 1.59 hours of processed speech is a pilot proof-of-concept for single-speaker VITS.
3. **No human evaluation:** Only automated MOS was computed.
4. **Pilot scope:** Proof-of-concept for multilingual phoneme clustering across Indic languages.

Reproducibility

All code, configurations, and manifests required to reproduce this experiment are available in the repository:

Component	Path
Data download	<code>tts/tts_data_download.py --lang gu</code>
Audio preprocessing	<code>tts/tts_audio_preprocess.py --lang gu</code>
G2P labeling	<code>tts/tts_g2p_labeling.py --lang gu</code>
Tokenizer	<code>tts/vits_tokenizer.py</code>
Data splitting	<code>tts/tts_split_data.py --lang gu</code>
VITS training	<code>tts/tts_vits_training.py --lang gu --model both</code>
Inference	<code>tts/tts_inference.py --lang gu</code>
Automated evaluation	<code>tts/tts_automated_eval.py --lang gu</code>
Manifest	<code>data/tts_gujarati_female/manifest.csv</code>
Phoneme vocab	<code>g2p/phoneme_vocab.json</code>
Cluster mapping	<code>g2p/phoneme_cluster_mapping.json</code>
Training log	<code>/mnt/d/tts_models/gu/training_log.json</code>
Automated MOS results	<code>results/tts_gujarati_female/automated_mos.csv</code>
Automated MOS report	<code>results/tts_gujarati_female/automated_mos_report.md</code>

Raw audio, processed audio, model checkpoints, and generated WAV samples are excluded from Git via `.gitignore`.